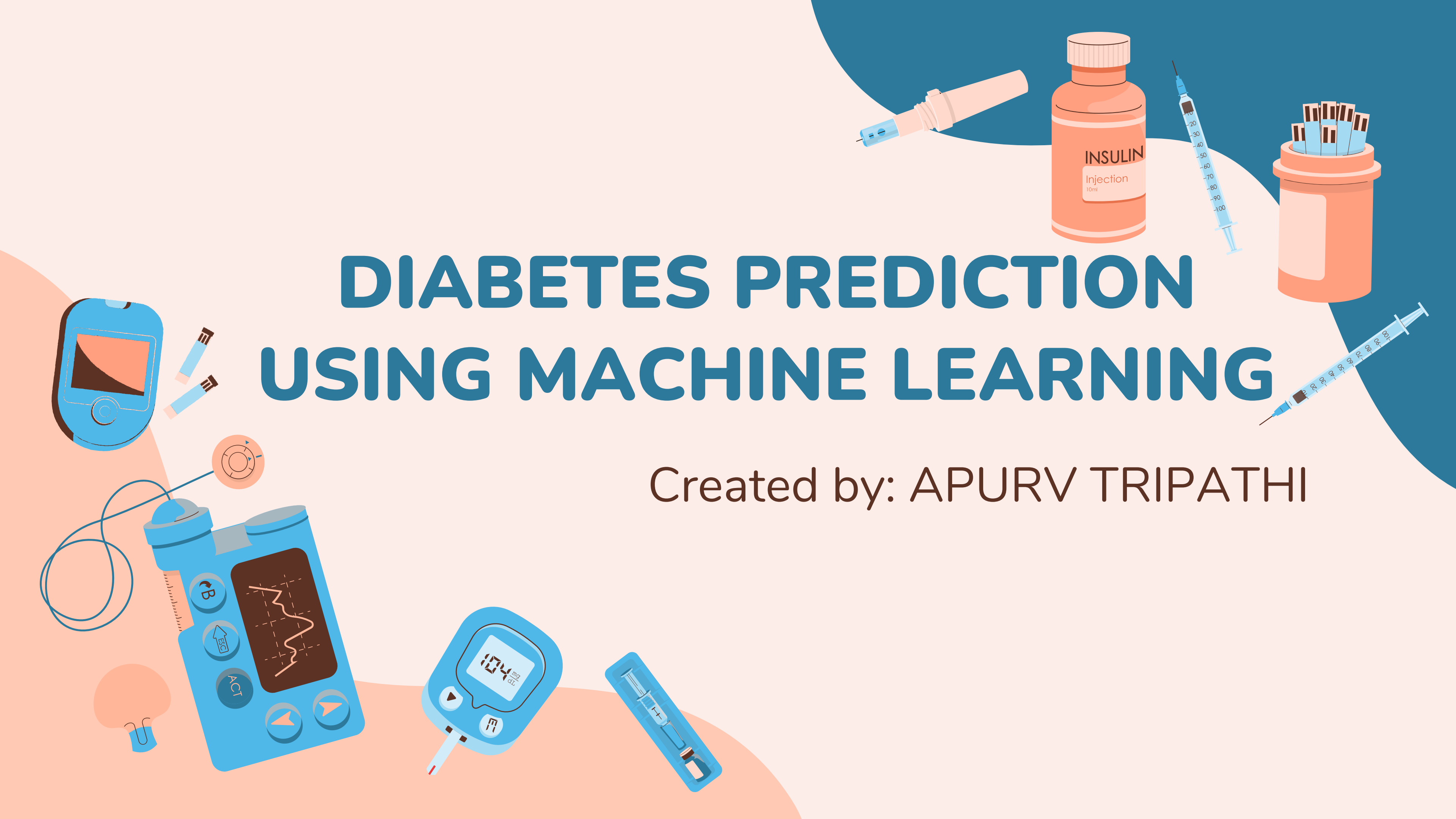


The background is a stylized medical illustration. It features a large orange pill bottle labeled 'INSULIN Injection 10ml' in the upper right. Next to it is a syringe with a blue plunger and needle. Below the bottle is a container filled with several insulin pens. In the lower left, there is a blue glucose meter with a screen showing a line graph and several buttons labeled 'B', 'ESC', 'ACT', and two arrow buttons. A blue cable is connected to the meter. To the right of the meter is a small blue device with a screen showing '104 mg/dL'. In the bottom center, there is a small blue insulin pen. The entire scene is set against a background of light orange and dark blue curved shapes.

# DIABETES PREDICTION USING MACHINE LEARNING

Created by: APURV TRIPATHI

# INTRODUCTION

- What is Diabetes?
  - A chronic disease that affects how your body processes blood sugar (glucose).
  - Can lead to serious health complications if not managed properly.
- Why Predict Diabetes?
  - Early detection can help in managing the disease effectively.
  - Machine learning can provide accurate predictions based on patient data.



# ABOUT THE COMPANY

Exposys Data Labs aims to Solve real world business problems like Automation, Big Data and data Science. our core team of experts in various technologies help businesses to identify issues,opportunities and prototype solutions using trending technologies like AI, ML, Deep Learning and Data Science. we follow a human-focussed and not technology driven approach to achieve success in our clients endeavours.

“Our discoveries are beyond belief and if you’re with us, you’ll discover a newer way to think!”

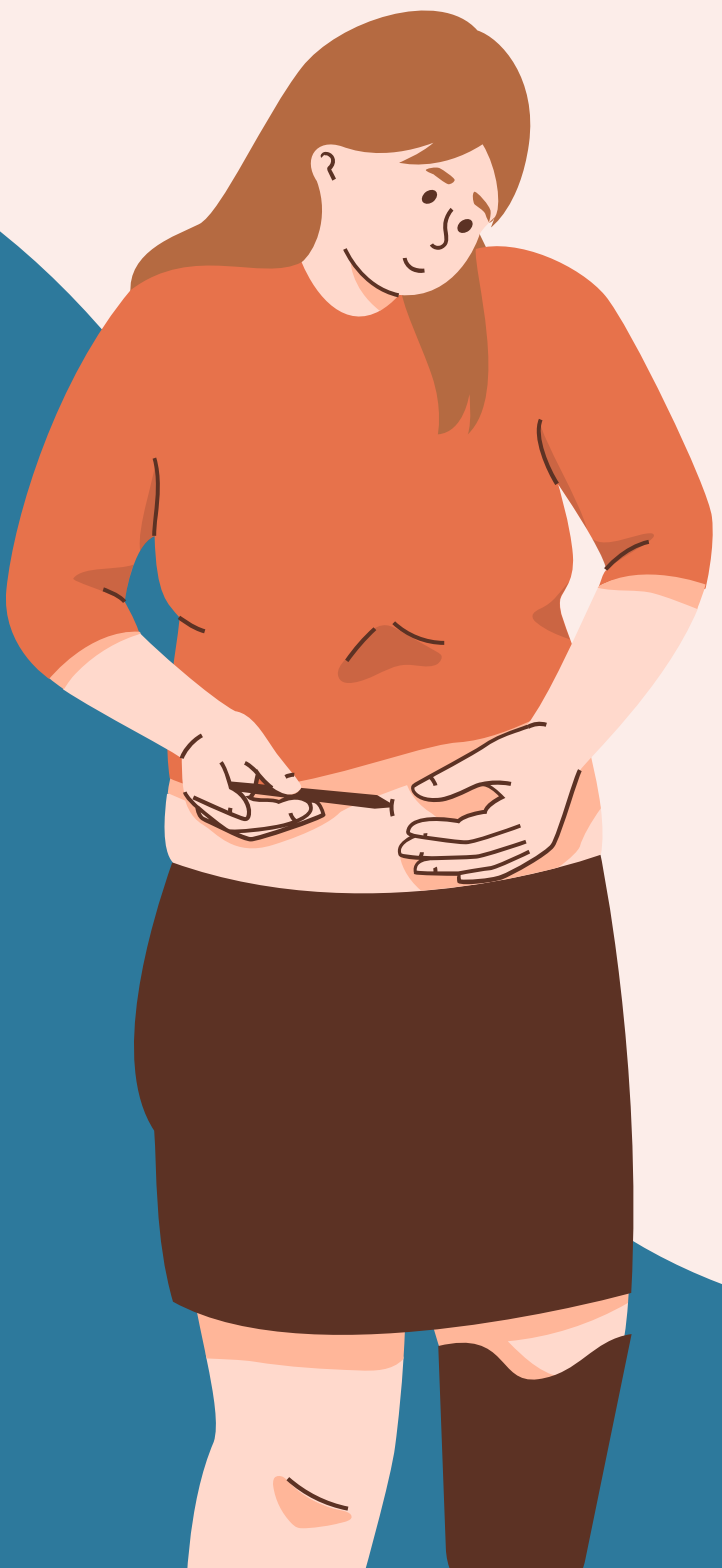
# PROJECT OVERVIEW

## Objective:

- To build a machine learning model that predicts the likelihood of diabetes based on patient data

## Dataset:

- Pima Indians Diabetes Dataset from Kaggle.
- Contains 768 samples with 8 features (e.g., Glucose, BMI, Age).

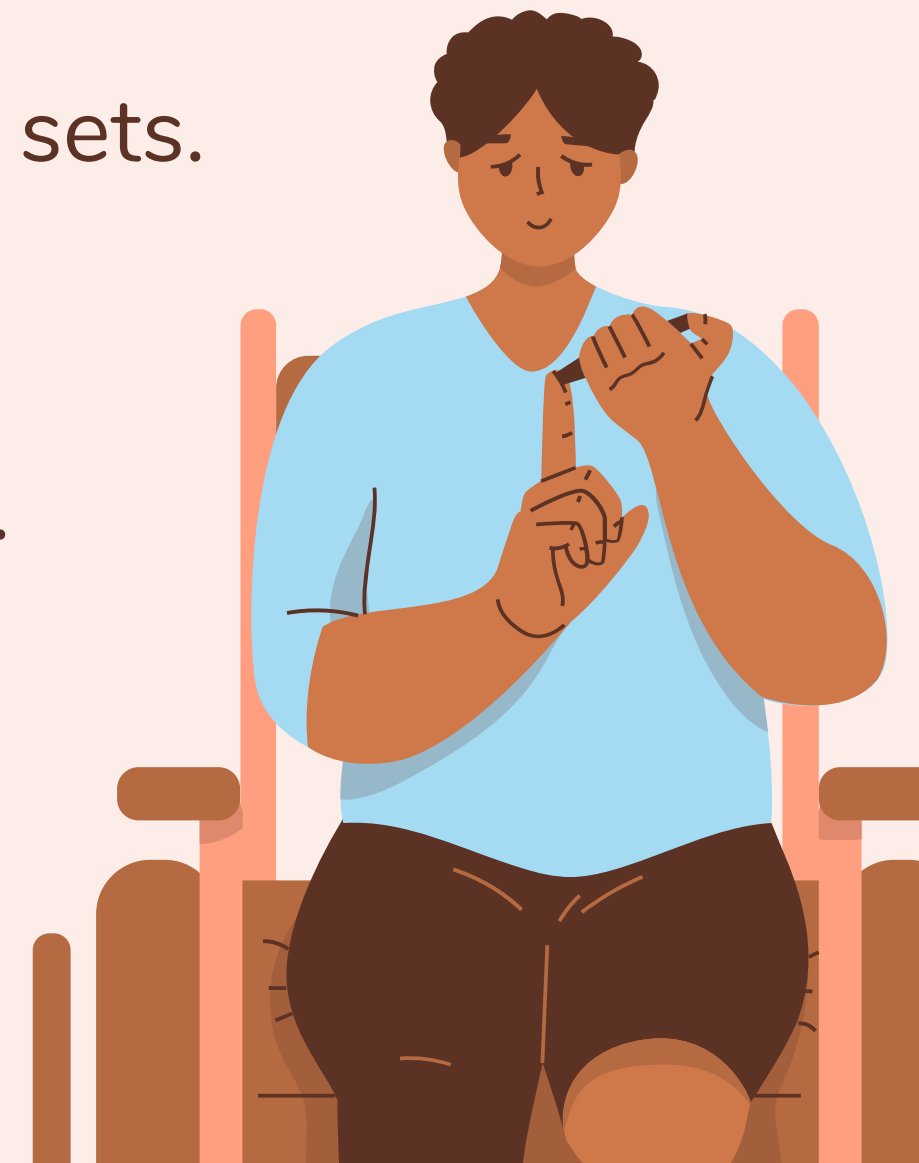


# DATA EXPLORATION

- Key Steps:
  - Loaded the dataset and checked for missing values.
  - Analyzed the distribution of features using histograms.
  - Visualized correlations between features using a heatmap.
- Insights:
  - No missing values, but some features had 0s (e.g., Glucose, BMI) that were replaced with the mean.
  - Strong correlation between Glucose and Outcome.

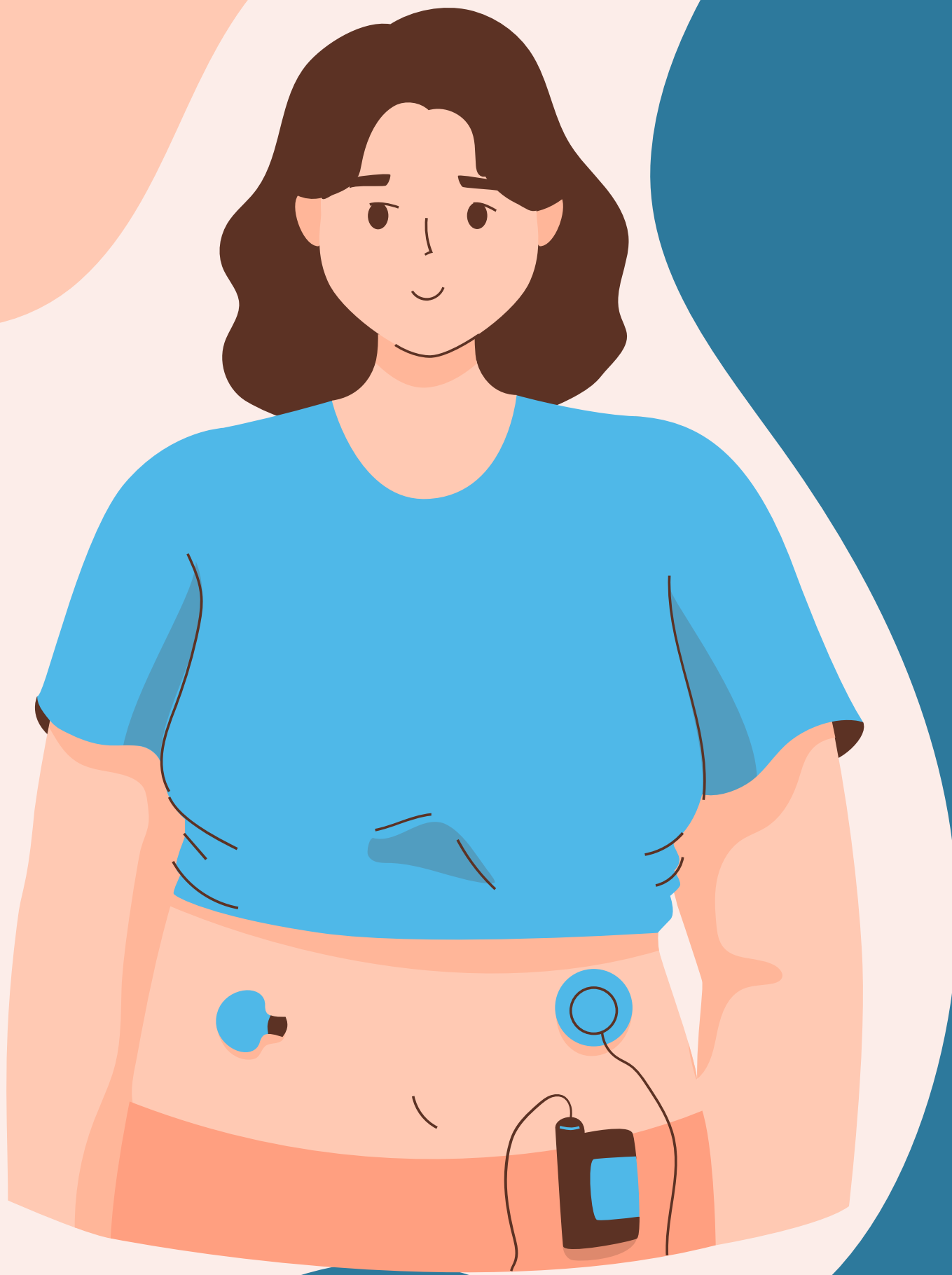
# Data Preprocessing

- Steps:
  - Replaced 0s with the mean for specific columns (e.g., Glucose, BMI).
  - Split the data into training (80%) and testing (20%) sets.
  - Scaled the features using StandardScaler.
- Why Scaling?
  - Ensures all features contribute equally to the model.



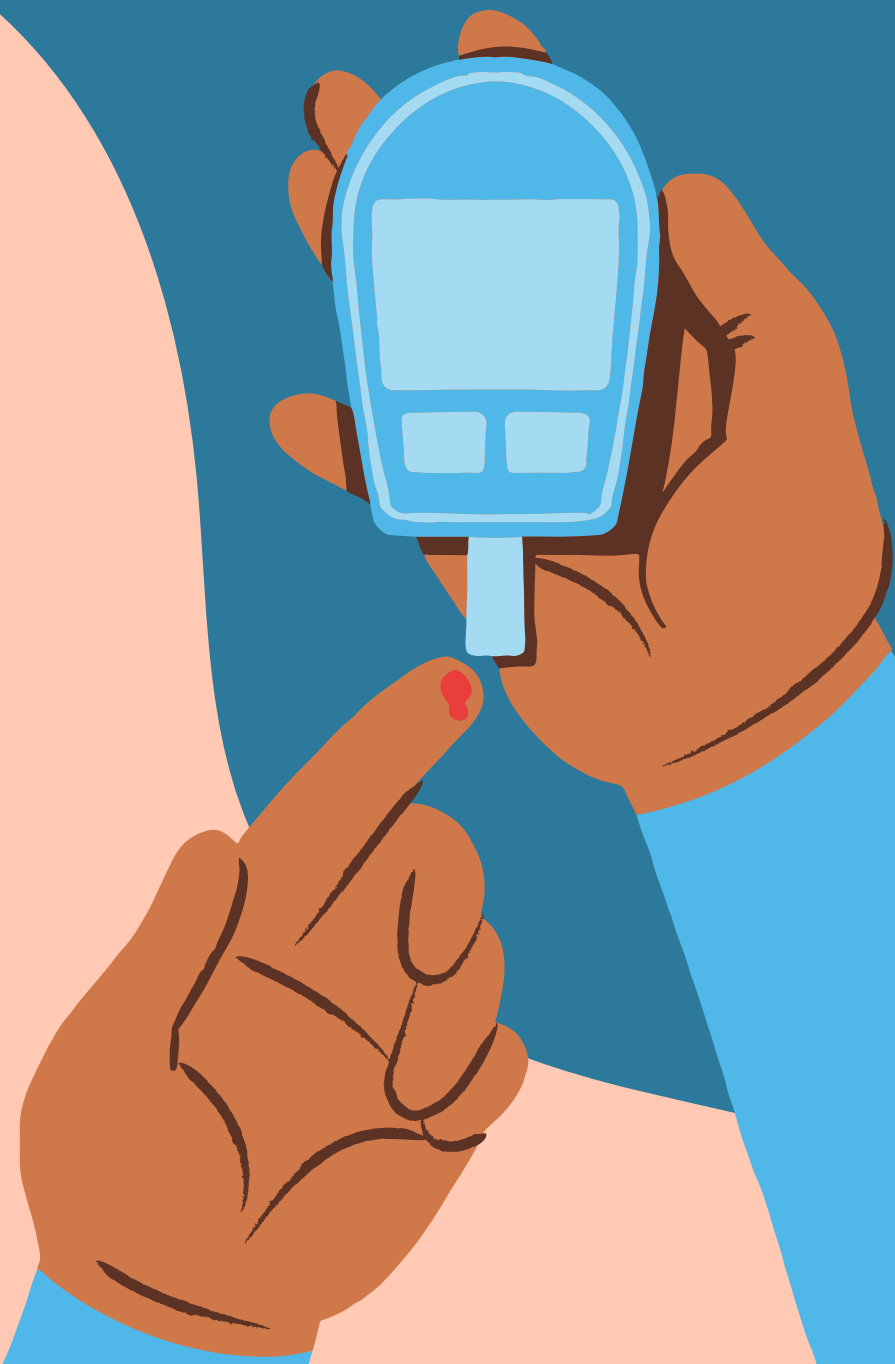
# MODEL BUILDING

- Algorithms Used:
  - Logistic Regression
  - Support Vector Machine (SVM)
  - Random Forest
  - K-Nearest Neighbors (KNN)
  - Naive Bayes
- Evaluation Metric:
  - Accuracy: Percentage of correct prediction



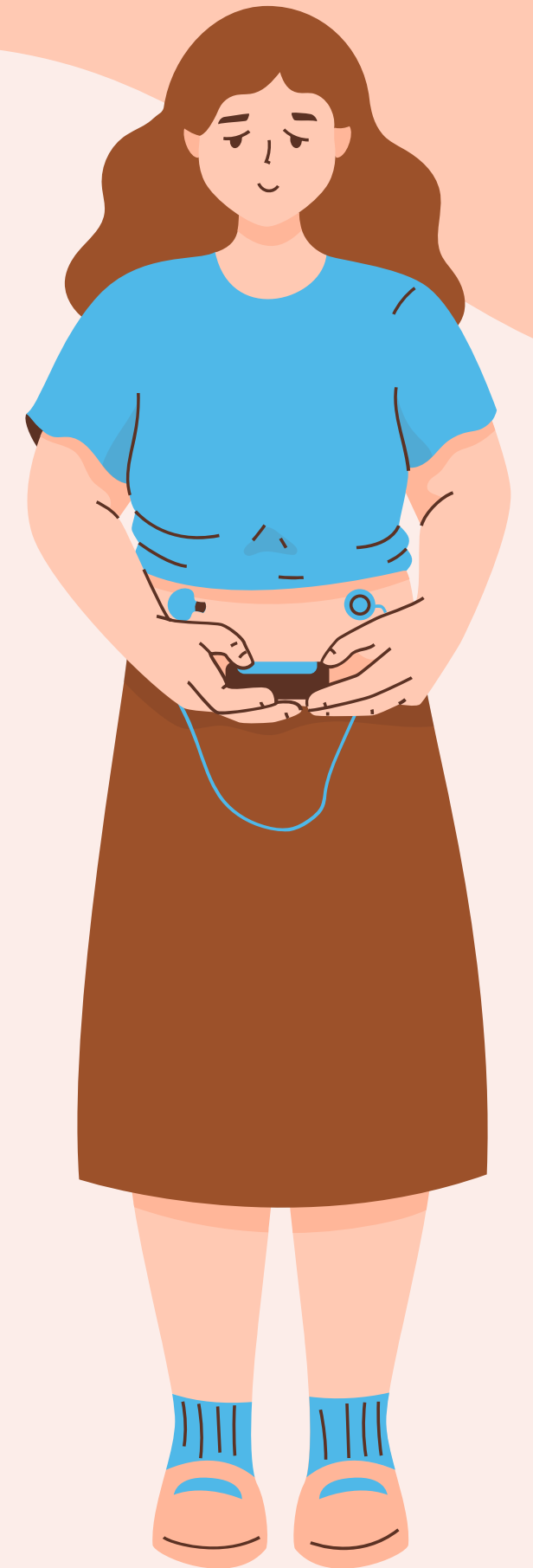
# MODEL COMPARISON

- Results:
  - Random Forest had the highest accuracy (e.g., 78%).
  - Logistic Regression and SVM also performed well.
- Why Random Forest?
  - Handles non-linear relationships and avoids overfitting.



# GUI FOR PREDICTIONS

- What is the GUI?
  - A user-friendly interface to input patient data and get predictions.
- Features:
  - Input fields for all 8 features (e.g., Glucose, BMI, Age).
  - A "Predict" button to display the result



# DEMONSTRATION

- Live Demo:
  - Show the GUI in action.
  - Input sample data and display the prediction result.



# CHALLENGES FACED

- Data Quality:
  - Handling 0s in features like Glucose and BMI.
- Model Selection:
  - Choosing the best algorithm for the dataset.
- GUI Development:
  - Integrating the model with a user-friendly interface



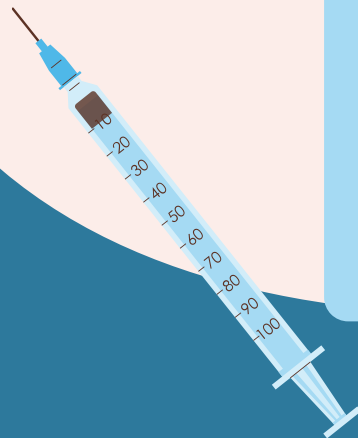
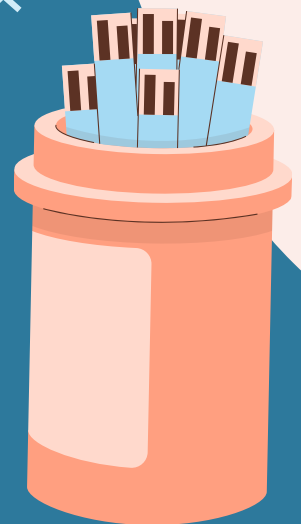
# FUTURE WORK

- Improvements:
  - Use a larger and more diverse dataset.
  - Experiment with deep learning models (e.g., Neural Networks).
  - Deploy the model as a web or mobile app.
- Applications:
  - Use in hospitals for early diabetes detection.
  - Integrate with wearable devices for real-time monitoring



# CONCLUSION

- Key Takeaways:
  - Built a machine learning model to predict diabetes with good accuracy.
  - Created a user-friendly GUI for easy predictions.
  - Demonstrated the potential of machine learning in healthcare.





# THANK YOU!

Thank you so much for watching our presentation! Do you have any questions, comments, or suggestions?

# GET IN TOUCH

## EMAIL ADDRESS:

apurvtriphathi425@gmail.com

## SOCIAL MEDIA:

ig: apurv\_vv\_007